NeurIPS 2026 Workshop on Explainability, Robustness, and Controllability of Language Modeling for Speech and Audio

Abstract

Large Language Models, including their multimodal variants have dominated the machine learning field for the last few years. In speech and audio, impressive results have been obtained in domains of speech/audio understanding and generation using language models. Specifically, in this workshop our goal is to invite contributions on i) Text-output speech/audio language models (LALMs) ii) Acoustic speech/audio language models iii) Audio representation leaning (E.g. audio tokenizers, self-supervised learning). We are particularly interested in questions around explainability, robustness and controllability. Given the high level of interest present on the speech/audio community around these topics, we expect this workshop to attract a large number of researchers.

Main Proposal

Introduction Recent years have witnessed a paradigm shift in machine learning, driven by the remarkable success of Large Language Models (LLMs) across diverse applications. Beginning with the introduction of the Transformer architecture [1], the field has evolved rapidly, with models like GPT-2 [2] and subsequent variants demonstrating unprecedented capabilities in natural language understanding and generation. This success has catalyzed a broader movement towards increasingly capable multimodal large language models [3], extending the reach of language modeling beyond traditional text-based tasks.

The expansion of LLMs into multimodal domains represents a natural and powerful evolution of this approach. Models such as GPT-4V [4], Gemini [5], and LLaVA [6] have demonstrated that language model architectures can effectively integrate and reason over multiple modalities, including images, video, and audio. This multimodal capability has opened new frontiers for machine learning applications and has begun reshaping how the research community approaches complex perception and reasoning tasks.

The speech and audio processing community has particularly been impacted from this shift. Recent developments have shown that language model frameworks can be successfully adapted for speech and audio understanding and generation tasks [7]. Significant progress has been achieved through powerful representation learning approaches based on self-supervised learning [8, 9] and discrete tokenization approaches [10], which enable the application of autoregressive language modeling techniques to acoustic signals. Models such as CLAP [11] also learn language and audio cross modal representations through contrastive learning.

Furthermore, generative models for audio have advanced substantially through language model-based approaches. Speech synthesis systems leveraging language model paradigms [12, 13] have achieved remarkable naturalness and controllability. Audio generation models [14, 15, 16, 17, 18, 19] have extended these capabilities to broader acoustic domains beyond speech. These successes have created substantial momentum and interest within the community around language modeling approaches for speech, music and audio.

This workshop aims to serve as a focal point for research and discussion at the intersection of language modeling and speech/audio processing. Our scope centers on acoustic language models, which can generally be classified into two main categories: (i) *generative acoustic LMs*, which are conditioned on preceding audio or text and generate audio or text [20, 21, 22]; and (ii) *audio-aware*

LMs, which are conditioned on audio or text and generate text [23, 24, 25]. The generative category can in turn be organized by the underlying generation approach, which broadly falls into two families: diffusion-based methods [16, 26, 27] and language-model-based methods [17, 18, 28]. Alongside these two categories, iii) we encourage work on audio representation learning that supports both of them, including audio tokenizers and self-supervised models. We further encourage contributions on methods that improve the interpretability and explainability of these models, as well as approaches for controllability. Given the strong momentum and broad applicability of these topics, we anticipate significant interest from the speech, audio, and machine learning research communities.

Scope and Topics of Interest This workshop covers the intersection of speech and audio processing with language modeling. We welcome submissions in three main categories:

- **Explainability.** We seek methods and studies that help explain how audio and speech language models make decisions. The goal is to show *why* a model produces a given output. This can be done through post-hoc interpretability, such as chain-of-thought prompting [29, 23] or attribution and explanation methods [30, 31]. It can also be done with models that are interpretable by design [25]. Another key topic is *faithfulness*: whether a model's stated reasoning matches what the model actually computes[32, 33].
- **Robustness.** We invite methods and studies on the robustness of audio and speech language models. We are interested in how models handle different factors, including but not limited to robustness against noise, adversarial attacks and prompt injections [32], hallucinations, and temporal shifts . This topic also covers the security and safety of these models, such as their behavior under distribution shift and their resistance to malicious inputs [34].
- **Controllability.** This area covers architectures and representations that make it easier to control specific parts of generated or edited audio. This includes changing speaker identity or prosody in speech, or rhythm and instrumentation in music. It also includes instruction-based generative models, such as expressive text-to-speech [35, 36]. Another important example application of controllability is audio editing [37, 38, 39], where we aim to control an aspect specified with a prompt and leave the rest intact.

Call for Papers We will publish a call for papers on audio language models revolving around the introduced themes. We will accept submissions with four pages in the NeurIPS format, and we will also allow an appendix. Submissions will be double blind and will be peer reviewed on OpenReview. We will do active publicity on Machine learning mailing groups (e.g. ML-News), social media (e.g. Linkedin, X, Bluesky), and instituional slack channels, and mailing groups. The timeline for paper reviewing will be as follows:

- August 29, 2026, AoE: Paper submission deadline (NeurIPS suggested deadline);
- September 19, 2026 AoE: Deadline for reviews.
- September 29, 2026, AoE: Paper decisions are announced (Mandatory NeurIPS notification date);

Note that we have put ten days between the deadline for reviews and the paper notification deadline, in order to allow for emergency reviews, in case needed.

Tentative Schedule The workshop will include four invited talks, a panel discussion, two poster sessions and oral presentations for paper submissions.

1. Opening remarks (9.00-9.15);
2. Plenary speaker 1 (9.15-10.00);
3. Two oral presentations from contributing authors (10.00-10.30);
4. Plenary speaker 2 (10.30-11.15);
5. Poster session and lunch break (11.15-12.45);
6. Plenary speaker 3 (12.45-13.30);
7. Two oral presentations from contributing authors (13.30-14.00);

8. Plenary speaker 4 (14.00-14.45);
9. Coffee break (14.45-15.15);
10. Plenary speaker 5 (15.15-16.00);
11. Poster session (16.00-16.30);
12. Panel discussion (16.30-17.30);
13. Closing remarks (17.30-17.45).

Note on the panel discussion

For the panel discussion we will have all of the invited speakers, and we will guide the discussion around the main questions we listed in the beginning of this document. Moreover, we will accept questions from the audience through a platform such as slido, in order to initiate discussions that involve the audience.

Attendence We expect around 50 to 100 in-person attendees. We would like to keep a relatively small group presentation format in order to create an interactive workshop that promotes discussions between the speakers and the audience.

Invited speakers and panelists:

- Prof. Wenwu Wang (University of Surrey), **Confirmed**
  Website: `https://www.surrey.ac.uk/people/wenwu-wang`
- Dr. Nick J. Bryan (Adobe Research), **Tentatively Confirmed**
  Website: `https://njb.github.io/index.html`
- Dr. Alexandre Defossez (Gradium), **Confirmed**
  Website: `https://ai.honu.io/`
- Prof. Dinesh Minosha (University of Maryland) **Confirmed**
  Website: `https://www.cs.umd.edu/people/dmanocha`
- Prof. Dilek Hakkani-Tur (University of Illinois at Urbana-Champaign) **Confirmed**
  Website: `https://siebelschool.illinois.edu/about/people/all-faculty/dilek`

We have contacted the researchers indicated above via email, and obtained their confirmation. Note that we will have five talks, each lasting around 35 minutes, with an additional ten minutes allocated for questions, in order to leave sufficient time for discussions. These plenary talks will be given by senior invitees of various backgrounds such as Prof. Dilek Hakkani Tur (Conversational LLM Agents), Prof. Dinesh Minosha (Audio Language Models), Prof. Wenwu Wang (Audio Generative Models, Langugage Modeling) Dr. Alexandre Defossez (Audio Tokenizers, representation learning) and Dr. Nick Bryan (Audio GenAI, Audio Editing in industrial setting). In case any of these researchers are unavailable for an unforeseen reason, we are confident that we will able to substitute them quickly as we have a large network from our several years in the AudioML community.

Diversity Statement We have organizers with varying levels of seniority. This includes a full-professor, an associate professor, and an assistant professor, two senior PhD students, and a junior PhD student. In terms of geographic diversity of the organizers, we have one professor from Europe (Ricard Marxer-France),

We have invited speakers that cover different targeted domains, areas of expertise, institutions, geographic locations (Europe-2, North America-3), industry (2) vs academia (3), genders and seniority.

Venue Preference We would prefer the workshop to be held at Atlanta, however we are also open to host it in Paris or Sydney if the logistical requirements favor the former two options.

Organizers

Organization Team

- **Francesco Paissan** PhD Student at Université Laval and Mila-Québec AI Insitute
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  Research Interests: Explainable Machine Learning, Audio Tokenization, Machine Learning for Speech and Audio, On-device Machine Learning
  Prior Scientific Challenge Organization Experience: Coordinator of the DCASE "Low-Complexity Acoustic Scene Classification" task 2022, ICASSP 2024 Workshop on Explainable AI for Speech Audio, Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio
- **Pooneh Mousavi** PhD student at Concordia/Mila University
  Website: `https://poonehmousavi.github.io/`
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  Research Interests: Audio Representation Learning, Audio Tokenization, Explainable Machine Learning
  Prior Workshop Organization Experience: Conversational AI Reading Group (2024- 2026)
- **Aravind Krishnan** PhD student at Saarland University
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  Research Interests: Machine Learning for Speech and Audio, Audio Tokenization, Explainable Machine Learning
  Prior Workshop Organization Experience: Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio
- **Ricard Marxer** Full Prof. at Université de Toulon Computer Science Dept., Visiting Prof. at ÉTS, researcher at ILLS, CNRS and Mila-Québec AI Institute
  Website: `https://www.ricardmarxer.com/`
  Email: `ricard.marxer@lis-lab.fr`
  Research Interests: Machine Learning for Speech and Audio, Audio Tokenization, Explainable Machine Learning, Representation Learning
  Prior Workshop Organization Experience: Coordinator of Dagstuhl Seminar 2016 on vocal interactivity in-and-between animal, humans and robots (VIHAR); CHiME workshop 2016; VIHAR workshops (2017, 2019, 2021, 2024); special sessions for Interspeech 2016, IEEE IJCNN 2020, EUSIPCO 2024, IEEE IJCNN 2026
- **Mirco Ravanelli** Associate Prof. at Concordia University Computer Science and Software Engineering Dept., Adjunct Prof. at Université de Montréal, Associate Academic Member Mila-Québec AI Institute
  Website: `https://sites.google.com/site/mircoravanelli/`
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  Research Interests: Conversational AI, Machine Learning for Speech and Audio, Audio Tokenization, Explainable Machine Learning, Speech Recognition
  Prior Workshop Organization Experience: NIPS 2018 Workshop on Interpretability and Robustness in Audio, Speech and Language, ICML 2020 Workshop on Self-Supervised Learning for Speech and Audio, Interspeech SpeechBrain Summit 2023, ICASSP 2024 Workshop on Explainable AI for Speech Audio, General Chair for ANNPR 2024, General Chair for 2025 IEEE Machine Learning for Signal Processing Conference
- **Cem Subakan** Assistant Prof. at Laval University Computer Science and Software Engineering Dept., Affiliate Assistant Prof. at Concordia University, Associate Academic Member at Mila-Québec AI Institute,
  Website: `ycemsubakan.github.io`
  Email: `cem.subakan@ift.ulaval.ca`
  Research Interests: Explainable Machine Learning, Machine Learning for Speech and Audio, Multimodal LLMs, Multi-Modal Representation Learning, Continual Learning
  Prior Workshop Organization Experience: Interspeech SpeechBrain Summit 2023, ICASSP 2024 Workshop on Explainable AI for Speech Audio, Sponsorship Chair for ANNPR 2024,

Interspeech 2025 Special Session on Explainable Machine for Speech and Audio, Lead General Chair for 2025 IEEE Machine Learning for Signal Processing Conference

Program Committee Paper evaluation will be divided among organizers and members of the following reviewer pool:

- **Luca Della Libera** PhD student at Mila + Concordia University. **Confirmed**
- **Emanuele Marconato** PhD student at University of Trento. **Confirmed**
- **David Beauchemin** PhD student at Université Laval. **Confirmed**
- **Steve Azzolin** PhD student at University of Trento. **Tentative**
- **Arsenii Gorin** Applied Scientist, Tali AI **Confirmed**
- **Zhepei Wang** Research Scientist at Adobe Research. **Confirmed**
- **Efthymios Tzinis** Research Scientist Google. **Tentative**
- **Ertug Karamatli** Independent Researcher **Tentative**
- **Benjamin Leblanc** PhD student at Université Laval. **Confirmed**
- **Antonio Longa** PhD student at University of Trento. **Tentative**
- **Martin Kocour** PhD student at Brno University of Technology. **Confirmed**
- **Honza Svec** PhD student Brno University of Technology. **Tentative**
- **Salah Zaiem** Research Scientist at Google Deepmind **Confirmed**
- **Firat Oncel** PhD student Concordia University. **Tentative**
- **Xuechen Liu** PhD student, University of Eastern Finland & Inria. **Confirmed**
- **Mirko Bronzi** Senior Research Scientist, Law Zero. **Confirmed**
- **Jonah Casebeer** Research Scientist Adobe, PhD Student University of Illinois at Urbana-Champaign **Confirmed**
- **Krishna Subramani** Research Engineer at Apple **Confirmed**
- **Eleonora Mancini** Research Scientist at Deezer **Confirmed**
- **Ada Tur** PhD Student at UIUC **Tentative**
- **Leda Sari** Research Scientist at Otter AI **Tentative**
- **Paris Smaragdis** Full Prof. at MIT EECS + MTA departments **Tentative**
- **Pascal Germain** Associate Prof. at CS department, Laval University **Tentative**
- **Dimitris Bralios** PhD Student at UIUC **Tentative**
- **Jihoon Jeong** PhD Student at Mila + Laval University **Confirmed**
- **Shubham Gupta** PhD Student at Mila + Laval University **Confirmed**

Additionally, if the number of submissions exceed the maximum of three papers per reviewer, we will recruit more reviewers from our access to a large pool of reviewers from our earlier organization experiences (e.g. IEEE MLSP 2025 conference (400+ reviewers), or ANNPR 2024, 2026), or XAI-SA ICASSP 2024 workshop. If even in this case the number of reviews is not high enough to handle the amount of submissions, we will recruit more reviewers from the pool of authors who submit to our workshop.

Note that as organizers we will review papers (3 papers each) as well. The organizers will also share the meta-reviewing duties by equally distributing the papers among themselves.

As indicated above, we have 32 designated reviewers (26 reviewers + 6 organizers). We will avoid conflict of interests, by adding more reviewers, if needed. Realistically, we foresee that we can have up to 100 reviewers without relying on authors that submitted to our workshop. This means that we would be able to review up-to around 100 submissions (not including author reviewers). Note that we will provide three reviews for each paper, at a minimum. We will use openreview, as we have extensive experience using it for IEEE MLSP 2025, ICASSP XAI-SA, and ANNPR 2024.

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